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Electronic device and method for wireless communication, and computer readable storage medium

Abstract

Provided in the present disclosure are an electronic device and method for wireless communication, and a computer readable storage medium. The electronic device comprises: a processing circuit, which is configured to: determine whether a target user equipment (UE) will apply a large intelligent surface (LIS)-assisted communication mode; and, if it is determined that the target UE will apply the LIS-assisted communication mode, send to the target UE configuration information of an LIS reference signal for channel state measurement in the LIS-assisted communication mode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) The present application is based on PCT filing PCT/CN2021/111658, filed Aug. 10, 2021, which claims priority to Chinese Patent Application No. 202010824989.3, titled "ELECTRONIC DEVICE AND METHOD FOR WIRELESS COMMUNICATION AND COMPUTER-READABLE STORAGE MEDIUM", filed on Aug. 17, 2020 with the China National Intellectual Property Administration, the entire contents of which are incorporated herein by reference in their entirety.
TECHNICAL FIELD
(2) The present disclosure relates to the field of wireless communication technology, in particular to the determination of a large intelligent surface (LIS)-assisted wireless communication mode. More particularly, the present disclosure relates to an electronic apparatus and a method for wireless communications, and a computer-readable storage medium.
BACKGROUND
(3) One of the main challenges of the next generation wireless network is to meet the growing demand for higher data rates. Meanwhile, another concern is the energy efficiency issues caused by the significant increase in the number of cellular connectivity devices. LIS has become a promising cost-effective technology to improve the spectrum and energy efficiency of a future wireless network. The LIS is a hyper surface composed of many small passive reflectors, and can modify an incident signal and guide a reflected wave to move in any predetermined direction, so as to obtain an ideal electromagnetic propagation environment with limited power consumption. For example, under the control of a base station, the LIS modifies a phase of an incident wave to obtain a reflected wave in an appropriate reflection direction to improve the signal quality of a receiver.
(4) FIG. **1** is a schematic diagram showing an LIS-based assisted communication mode. Due to the existence of an obstacle or other reasons, the communication quality of a direct link from gNB to user equipment (UE) served by gNB is poor. In this case, the transmission mode of the UE may be changed to use the LIS to assist communication. As shown in FIG. **1**, a signal for a user is reflected by the LIS and then provided to the user in a certain reflection direction.
(5) However, after the introduction of the LIS into a cellular communication system, the switching of the user transmission mode not only affects the link quality of the user itself, but also changes the channel quality of other users due to the passive reflection property of the LIS. Therefore, it is desirable to provide a communication mechanism in which the transmission modes of multiple users are considered cooperatively.
SUMMARY OF THE INVENTION

(6) In the following, an overview of the present disclosure is given simply to provide basic understanding to some aspects of the present disclosure. It should be understood that this overview is not an exhaustive overview of the present disclosure. It is not intended to determine a critical part or an important part of the present disclosure, nor to limit the scope of the present disclosure. An object of the overview is only to give some concepts in a simplified manner, which serves as a preface of a more detailed description described later.
(7) According to an aspect of the present disclosure, an electronic apparatus for wireless communications is provided. The electronic apparatus includes processing circuitry. The processing circuitry is configured to: determine whether target user equipment is to apply an LIS-assisted communication mode; and in a case of determining the target user equipment is to apply the LIS-assisted communication mode, transmit, to the target user equipment, configuration information of an LIS reference signal to be used for channel status measurement under the LIS-assisted communication mode.
(8) According to another aspect of the present disclosure, a method for wireless communications is provided. The method includes: determining whether target user equipment is to apply an LIS-assisted communication mode; and in a case of determining the target user equipment is to apply the LIS-assisted communication mode, transmitting, to the target user equipment, configuration information of an LIS reference signal to be used for channel status measurement under the LIS-assisted communication mode.
(9) According to an aspect of the present disclosure, an electronic apparatus for wireless communications is provided. The electronic apparatus includes processing circuitry. The processing circuitry is configured to: acquire, from a base station, configuration information of an LIS reference signal to be used for channel status measurement under an LIS-assisted communication mode; and perform measurement of the LIS reference signal based on the configuration information.
(10) According to another aspect of the present disclosure, a method for wireless communications is provided. The method includes: acquiring, from a base station, configuration information of an LIS reference signal to be used for channel status measurement under an LIS-assisted communication mode; and performing measurement of the LIS reference signal based on the configuration information.
(11) According to other aspects of the present disclosure, there are further provided computer program codes and computer program products for implementing the methods for wireless communications above, and a computer-readable storage medium having recorded thereon the computer program codes for implementing the methods for wireless communications described above.
(12) With the electronic apparatus and method according to the embodiment of the present disclosure, a specific LIS reference signal is configured for the base station and UE under the LIS-assisted communication mode, so as to achieve accurate channel measurement under the LIS-assisted communication mode to avoid or alleviate the interferences between UEs.
(13) These and other advantages of the present disclosure will be more apparent from the following detailed description of preferred embodiments of the present disclosure in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) To further set forth the above and other advantages and features of the present disclosure, detailed description will be made in the following taken in conjunction with accompanying drawings in which identical or like reference signs designate identical or like components. The

accompanying drawings, together with the detailed description below, are incorporated into and form a part of the specification. It should be noted that the accompanying drawings only illustrate, by way of example, typical embodiments of the present disclosure and should not be construed as a limitation to the scope of the disclosure. In the accompanying drawings:

(2) FIG. **1** is a schematic diagram showing an LIS-based assisted communication mode;

(3) FIG. **2** shows a schematic example of an application scenario of an LIS;

(4) FIG. **3** shows a possible communication mode in a case of considering two UEs and one LIS;

(5) FIG. **4** shows another possible communication mode in a case of considering two UEs and one LIS;

(6) FIG. **5** shows another possible communication mode in a case of considering two UEs and one LIS;

(7) FIG. **6** is a block diagram showing functional modules of an electronic apparatus for wireless communications according to an embodiment of the present disclosure;

(8) FIG. **7** shows an example of an LIS set and available serving UE set;

(9) FIG. **8** shows an example of a reflected beam direction of the LIS;

(10) FIG. **9** shows an example of LIS reference signals configured for two LISs respectively, in a case that the LIS reference signals are non-precoded/non-beamformed reference signals;

(11) FIG. **10** shows a transmission example of the LIS reference signals of FIG. **9**;

(12) FIG. **11** to FIG. **16** show other configuration examples of LIS reference signals of multiple LISs, in a case that the LIS reference signals are non-precoded/non-beamformed reference signals;

(13) FIG. **17** shows an example of LIS reference signals configured for three LISs respectively, in a case that the LIS reference signals are precoded/beamformed reference signals;

(14) FIG. **18** shows a transmission example of the LIS reference signals of FIG. **17**;

(15) FIG. **19** to FIG. **23** show other configuration example of LIS reference signals of multiple LISs, in a case that the LIS reference signals are precoded/beamformed reference signals;

(16) FIG. **24** shows a schematic diagram of a scenario of LIS 1 and LIS 2 as well as UEs within a coverage range thereof;

(17) FIG. **25** is a block diagram showing functional modules of an electronic apparatus for wireless communications according to an embodiment of the present disclosure;

(18) FIG. **26** is a schematic diagram showing an example of information procedure among a base station (gNB), LIS and UE;

(19) FIG. **27***a* shows a flowchart of a method for wireless communications according to an embodiment of the present disclosure;

(20) FIG. **27***b* shows a flowchart of a method for wireless communications according to another embodiment of the present disclosure;

(21) FIG. **28** is a block diagram showing a first example of an exemplary configuration of an eNB or gNB to which the technology of the present disclosure may be applied;

(22) FIG. **29** is a block diagram showing a second example of an exemplary configuration of an eNB or gNB to which the technology of the present disclosure may be applied;

(23) FIG. **30** is a block diagram showing an example of an exemplary configuration of a smartphone to which the technology according to the present disclosure may be applied;

(24) FIG. **31** is a block diagram showing an example of an exemplary configuration of a car navigation apparatus to which the technology according to the present disclosure may be applied; and

(25) FIG. **32** is a block diagram of an exemplary block diagram illustrating the structure of a general purpose personal computer capable of realizing the method and/or device and/or system according to the embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(26) An exemplary embodiment of the present disclosure will be described hereinafter in conjunction with the accompanying drawings. For the purpose of conciseness and clarity, not all

features of an embodiment are described in this specification. However, it should be understood that multiple decisions specific to the embodiment have to be made in a process of developing any such embodiment to realize a particular object of a developer, for example, conforming to those constraints related to a system and a service, and these constraints may change as the embodiments differs. Furthermore, it should also be understood that although the development work may be very complicated and time-consuming, for those skilled in the art benefiting from the present disclosure, such development work is only a routine task.

(27) Here, it should also be noted that in order to avoid obscuring the present disclosure due to unnecessary details, only a device structure and/or processing steps closely related to the solution according to the present disclosure are illustrated in the accompanying drawing, and other details having little relationship to the present disclosure are omitted.

First Embodiment

(28) FIG. **2** shows a schematic example of an application scenario of LIS. Multiple LISs are deployed in a cell, each LIS may serve one or more UEs, and each UE may also be served by one or more LISs. For example, in a case of considering two UEs and one LIS, there may be three modes: 1) gNB directly serves the two UEs without the assistance of the LIS, as shown in FIG. **3**; 2) the two UEs communicate with the assistance of the LIS, as shown in FIG. **4**; 3) Only UE2 communicates with the assistance of the LIS, as shown in FIG. **5**. It can be seen that in FIGS. **4** and **5**, the LIS-assisted communication for one UE will produces interferences on the other UEs due to the passive reflection property of the LIS. In a case of considering multiple LISs, the interferences condition is more complex.

(29) Therefore, it is necessary to consider a cooperative relationship among multiple LISs and multiple users. Therefore, it is desirable to provide a technology for dynamically selecting and switching the transmission modes of UE. In order to select an appropriate association mode between the LIS and the user and an appropriate reflecting beam direction of each LIS, it is necessary to accurately estimate channel status information of a channel among a base station, LIS and UE. With respect to, but not limited to, this purpose, an electronic apparatus **100** for wireless communications is provided according to this embodiment.

(30) FIG. **6** is a block diagram showing functional modules of an electronic apparatus **100** according to an embodiment of the present disclosure. The electronic apparatus **100** includes a determination unit **101** and a communication unit **102**. The determination unit **101** is configured to determine whether a target UE is to apply an LIS-assisted communication mode. The communication unit **102** is configured to: in a case that the determination unit **101** determines the target UE is to apply the LIS-assisted communication mode, transmit, to the target UE, configuration information of an LIS reference signal to be used for channel status measurement under the LIS-assisted communication mode.

(31) The determination unit **101** and the communication unit **102** may be implemented by one or more processing circuits. The processing circuits may be implemented as a chip and a processor, for example. Furthermore, it should be understood that functional units of the electronic apparatus shown in FIG. **6** are only logic modules divided according to the implemented specific functions, and are not intended to limit the specific implementations.

(32) The electronic apparatus **100** may be arranged on a base station side or communicatively connected to the base station, for example. The base station described herein may also be a Transmission Receive Point (TRP) or an Access Point (AP). Here, it should also be noted that the electronic apparatus **100** may be implemented in a chip level, or in an apparatus level. For example, the electronic apparatus **100** may function as the base station itself, and may also include an external device such as a memory, a transceiver (not shown). The memory may be used to store programs required by the base station to achieve various functions and related data information. The transceiver may include one or more communication interfaces to support communication with different devices (e.g., UE, other base stations or the like). Implementations of the transceiver are

not limited here.
(33) Since the LIS is a passive array, and does not emit a new signal by itself, in the LIS-assisted communication mode, the base station is required to assist the LIS in beam scanning for channel status measurement. An LIS reference signal is provided according to this embodiment to assist the LIS in beam scanning. With this LIS reference signal, UE may measure channel status information of a channel corresponding to a base station-LIS-UE for each beam from each LIS, and select an appropriate beam, namely the LIS and a reflection direction corresponding to the beam, for communication based on the measured channel status information.
(34) The target UE described here is the UE that may apply the LIS-assisted communication mode. For example, the determination unit **101** is configured to determine that the target UE is to apply the LIS-assisted communication mode in a case that communication quality of the target UE is lower than a predetermined threshold. The communication quality of the target UE may be measured by a quality of service (QOS) index such as reference signal reception power (RSRP), reference signal reception quality (RSRQ), signal to interference and noise ratio (SINR).
(35) For example, the communication unit **102** may acquire, from the target UE, communication quality information, such as a Channel Quality Indicator (CQI), of the target UE, or the determination unit **101** may estimate the communication quality of the target UE.
(36) In addition, the determination unit **101** may determine that the target UE is to apply the LIS-assisted communication mode in response to an LIS assisting request from the target UE. In this case, for example, when detecting that the communication quality of the target UE is lower than a predetermined threshold, the target UE transmits the LIS assisting request to the base station to request to switch to the LIS-assisted communication mode.
(37) When determining whether the target UE is to apply the LIS-assisted communication mode, the determination unit **101** may also take into account a priority level of the target UE. For example, when the priority level of the target UE is high, the possibility of applying the LIS-assisted communication mode is increased. The priority level information of the target UE may be acquired by the communication unit **102** from the target UE, or pre-stored on the base station side.
(38) The communication unit **102**, in response to the determination unit **101** determining that the target UE is to apply the LIS-assisted communication mode, transmits, to the target UE, configuration information of an LIS reference signal to be used for channel status measurement under the LIS-assisted communication mode.
(39) In an example, a mode of the LIS reference signal of each LIS may be predetermined. The communication unit **102** provides the UE with the predetermined configuration information of the LIS reference signal of each LIS.
(40) In another example, the configuration information of the LIS reference signal to be provided to the UE may be dynamically determined. For example, the determination unit **101** may also be configured to determine, in the case of determining the target UE is to apply the LIS-assisted communication mode, an LIS set to be used for LIS-assisted communication for the target UE and an available serving UE set for each LIS in the LIS set. For example, the LIS set includes respective LISs within whose coverage range the target UE is located, and the available serving UE set of an LIS includes UEs within a coverage range of the LIS.
(41) For ease of understanding, FIG. **7** shows an example of an LIS set and an available serving UE set. In the example of FIG. **7**, two LISs (LIS.sub.1 and LIS.sub.2) and several UEs are shown. UE k is the target UE, and a coverage range of the LIS is represented by a semicircle (a radius is Rn). It can be seen that the UE k is within the coverage range of LIS.sub.1 and LIS.sub.2, and thus the LIS set for the UE k is L.sub.k={LIS.sub.1, LIS.sub.2}. The servable UE set (i.e., the available serving UE set) within the coverage range of LIS.sub.1 is S.sub.1={UE 1, UE 2, UE 3, UE k}. Similarly, the available serving UE set for LIS.sub.2 is S.sub.2={UE 4, UE 5, UE k}. Therefore, the final available serving UE set is a union of S.sub.1 and S.sub.2, that is, S.sup.k=S.sub.1 ∪S.sub.2={UE 1, UE 2, UE 3, UE 4, UE 5, UE k}.

(42) In order to determine the LIS set and the available serving UE set, the communication unit **102** may also be configured to acquire, from the target UE, an identifier of the LIS within whose coverage range the target UE is located. On the other hand, since a position of the LIS is fixed, a position of the LIS may be pre-stored on the base station side, and the determination unit **101** may determine the LIS set and the available serving UE set based on pre-stored location information of the LIS and location information of each UE.
(43) The communication unit **102** is configured to transmit, to each of UE in the available UE set, configuration information of the LIS reference signal related to the LIS in the LIS set. The communication unit **102** may transmit the configuration information to the UE via radio resource control (RRC) signaling. This configuration information is used to tell the UE how to receive and parse the LIS reference signal, how to report a measurement result, etc., for example.
(44) The communication unit **102** transmits the LIS reference signal to each UE via a corresponding LIS according to the configuration information of the LIS reference signal, and acquires a measurement result reported by each UE for measuring the LIS reference signal.
(45) For example, the communication unit **102** may periodically transmit the LIS reference signal and periodically acquire the measurement result of the LIS reference signal.
(46) In addition, in a case that the configuration information indicates an aperiodic reporting of the measurement result, the communication unit **102** is also configured to transmit a trigger signaling for reporting the measurement result of the LIS reference signal to each UE, and acquire an aperiodic measurement result of the LIS reference signal.
(47) For example, the configuration information of the LIS reference signal may include one or more of the following: a time-frequency resource location where the LIS reference signal is located, a correspondence relationship between LIS reference signals on different time-frequency resource locations and LISs, periodical reporting or non-periodical reporting of a measurement result.
(48) With this configuration information, the UE may know on which time-frequency resources the LIS reference signal is received, which LIS the received LIS reference signal comes from, and further which reflecting beam direction of the LIS the received LIS reference signal comes from, and whether to perform periodical reporting or non-periodical reporting of a measurement result.
(49) In other words, the correspondence relationship between LIS reference signals on different time-frequency resource locations and LISs may also include a correspondence relationship between LIS reference signals on different time-frequency resource locations and reflecting beam directions of different LISs in more detail.
(50) FIG. **8** shows an example of a reflecting beam direction of LIS. A semicircular reflection range of an LIS is divided equally into 6 sectors. Each sector represents a reflecting beam direction, and may be marked with an ID (beam 1, beam 2 . . . beam 6 are marked in FIG. **8**).
(51) For example, the LIS reference signal may be configured as follows: LIS reference signals for different LISs are distinguished in frequency domain, and LIS reference signals for different reflecting beam directions of the same LIS are distinguished in time domain. That is, LIS reference signals for different LISs occupy different frequency domain resources, and LIS reference signals for different reflecting beam directions of the same LIS occupy different time domain parts of the frequency domain resources of the LIS.
(52) For example, the reference signals for different LISs are distributed on different subcarriers of orthogonal frequency division multiplexing (OFDM) symbols, and the LIS reference signals for different reflecting beam directions of the same LIS are distributed on different resource elements (RE) of the same subcarrier of the OFDM.
(53) In an example, the LIS reference signal is non-precoded/non-beamformed reference signal, and the LIS reference signals for different LISs do not overlap in time domain. In this case, since the LIS reference signal is non-directional and may be reflected by multiple LISs at the same time, the LIS reference signals for different LISs should be assigned with different durations, so that the

UE may distinguish the LIS reference signals from different LISs.
(54) FIG. **9** shows an example of LIS reference signals configured for two LISs respectively in a case that the LIS reference signal is non-precoded/non-beamformed reference signal. The 4th row of a subcarrier is allocated to LIS 1, and the 10th row is allocated to LIS 2. That is, the LIS reference signals for different LISs are distributed on different subcarriers of OFDM symbols, and RE occupied by the LIS reference signal in 4th row and RE occupied by the LIS reference signal in 10th row are interlaced in time domain. Different columns in each row are assigned to different reflecting beam directions of the corresponding LISs. Suppose an LIS has a reflecting beam direction distribution shown in FIG. **8**, for example, the 2th column in the 4th row is assigned to a beam 1 of LIS 1, the 3th column in the 10th row is assigned to a beam 2 of LIS 2, and so on.
(55) FIG. **10** shows a transmission example of the LIS reference signal of FIG. **9**. The LIS reference signal is first transmitted from a base station to a corresponding LIS, and then the LIS transmits a reflecting beam with a reflecting beam direction corresponding to the LIS reference signal to the UE. In the upper part of FIG. **10**, the transmission of LIS reference signal (LIS-RS) for LIS 1 is shown, where LIS ID=1 represents LIS 1, Beam ID=1 represents Beam 1, Subcarrier=4 represents that the LIS-RS is allocated with a subcarrier of the 4th row, and Type ID=2 represents that the reference signal is non-precoded/non-beamformed. In the lower part of FIG. **10**, the transmission of the LIS reference signal (LIS-RS) for LIS 2 is shown. Similarly, LIS ID=2 represents LIS 2, Beam ID=2 represents Beam 2, and Subcarrier=10 represents that the LIS-RS is assigned with a subcarrier of the 10th row. As shown in FIG. **10**, the transmission of LIS reference signals for LIS 1 and that for LIS 2 are separated in time.
(56) It should be understood that FIG. **9** is only a configuration example, rather than limitation, and any configuration mode, as long as it meets the above principles, is feasible. For ease of understanding, FIGS. **11** to **16** show another configuration examples of LIS reference signals of multiple LISs, in a case that the LIS reference signal is non-precoded/non-beamformed reference signal. In the examples of FIGS. **13**, **14** and **16**, each row represents less than 6 reflecting beam directions of a reference signal block, that is, it is impossible to scan all reflecting beam directions of an LIS at one time. In this case, two reference signal blocks may be transmitted consecutively to represent different reflecting beam directions in different time slots, so as to complete scanning of all reflecting beam directions.
(57) In another example, the LIS reference signal is a precoded/beamformed reference signal, and the LIS reference signals for different LISs overlap in the time domain. Because the LIS reference signal is digitally precoded, each OFDM symbol may have different subcarriers. The base station may transmit a corresponding LIS reference signal directionally according to a position of each LIS. The UE may distinguish LIS reference signals from different LISs according to the subcarriers. In other words, in this case, the LIS reference signals for different LISs may overlap in the time domain, thus having higher configuration flexibility.
(58) FIG. **17** shows an example of LIS reference signals configured for three LISs respectively in a case that the LIS reference signal is a precoded/beamformed reference signal. The 4th row of a subcarrier is allocated to LIS 1, the 7th row is allocated to LIS 2, and the 10th row is allocated to LIS 3. That is, the LIS reference signals for different LISs are distributed on different subcarriers of OFDM symbols, and RE occupied in each row overlaps in time domain. Different columns in each row are assigned to different reflecting beam directions of a corresponding LIS. Suppose LIS has the reflecting beam direction distribution shown in FIG. **8**, for example, the 4th column in the 4th, 7th and 10th rows is assigned to beam 1 of LIS 1, beam 1 of LIS 2 and beam 1 of LIS 3 respectively, and so on.
(59) FIG. **18** shows a transmission example of the LIS reference signals in FIG. **17**. Similarly, the LIS reference signal is first transmitted from a base station to a corresponding LIS, and then the LIS transmits the reflecting beam with the reflecting beam direction corresponding to the LIS reference signal to the UE. As shown in FIG. **18**, the base station simultaneously transmits LIS

reference signals for LIS 1, LIS 2 and LIS 3 on different subcarriers (subcarriers 4, 7 and 10), and the reflecting beam direction of the LIS 1, LIS 2 and LIS 3 each is beam 1. Type ID=1 represents that the reference signal is precoded/beamformed.
(60) It should be understood that FIG. **17** is only a configuration example, rather than limitation, any configuration mode, as long as it meets the above principles, is feasible. For ease of understanding, FIGS. **19** to **23** show another configuration example of LIS reference signals of multiple LISs in a case that the LIS reference signal is a precoded/beamformed reference signal. In the examples of FIG. **20** to FIG. **23**, the LIS reference signals of the multiple LISs partially overlap in the time domain.
(61) As described above, the communication unit **102** transmits the LIS reference signal according to the configuration information of the LIS reference signal, and meanwhile, the base station sequentially controls the reflecting beam direction of each LIS to obtain the desired reflecting beam for downlink beam scanning of the base station-LIS-UE. The UE sequentially measures received signal strength of a corresponding LIS reference signal, and feeds a measurement result back to the base station. The base station selects an appropriate channel for the UE based on the measurement result to perform LIS-assisted communication.
(62) As an example, the communication unit **102** may be configured to acquire, from the UE, with respect to each LIS in the LIS set, information of a reflecting beam direction in which the target UE receives signal of the maximum strength from the LIS.
(63) Specifically, the target UE measures the LIS reference signal in each reflecting beam direction of each LIS, and P.sub.n, k, i is used to represent a signal strength in the i-th reflecting beam direction of LIS n that the target UE k receives. Thus, the reflecting beam direction in which the target UE receives signal of the maximum strength from the LIS n i.sub.n,k,opt (i.e., an optimal reflecting beam direction) is determined by the following equation.
(64) $i_{n,k,\mathrm{opt}} = \mathrm{argmax}_{i=1,2,\ .Math.}\ \frac{P_{n,k,l}}{.Math._{k'=k} P_{n,k',l} + P_N}$ (1) where P.sub.N is the noise power, n is an ID of LIS in the LIS set L.sub.k, k is an ID of the target UE, and i is an ID of the reflecting beam.
(65) In addition, the communication unit **102** is also configured to acquire, from each UE among at least a part of remaining UEs in the available serving UE set other than the target UE, with respect to each LIS in the LIS set, information of a reflecting beam direction in which the UE receives signal of a strength meeting a communication quality requirement.
(66) The remaining UE k′ (k′≠k) in the available serving UE set S.sup.k may determine a set of the reflecting beam directions in which the communication quality requirement of the remaining UE k′ is met by the following equation.
(67) $i_{n,k'} = \{k'\ .Math.\ \frac{P_{n,k',t}}{.Math._{m\neq k'} P_{n,1} + P_N} > \mathrm{SINR}_{\min,k'}\}$ (2)
(68) i.sub.n,k′ represents a set of reflecting beam directions of LIS n in which the communication quality requirement of the UE k′ is met, P.sub.n,k′,t represents a signal strength for the UE k′ in the i-th reflecting beam direction of LIS n that the UE k′ receives, P.sub.n,m,i represents a signal strength for the remaining UE other than the UE k′ in the i-th reflecting beam direction of LIS n that the UE k′ receives, and SINR.sub.min,k′ represents a minimum value of SINR required to meet the communication quality requirement of the UE k′.
(69) If none of reflecting beams of an LIS can meet the communication quality requirement of the UE k′, the UE k′ may not report for the LIS. If none of reflecting beams of all of the LISs can meet the communication quality requirement of the UE k′, the UE k′ may continue to maintain direct communication with the base station without reporting the measurement result.
(70) For example, the measurement result reported by the UE may include an identifier (ID) of the LIS and an ID of the reflecting beam direction. For the target UE, the measurement result represents an optimal reflecting beam direction of the LIS in the LIS set. For the remaining UE, the

measurement result represents an LIS and a reflecting beam direction of the LIS that meet the communication quality requirement of the remaining UE.
(71) Alternatively, the measurement result reported by the UE may include a time-frequency resource location of the LIS reference signal. Since the time-frequency resource location of the LIS reference signal corresponds to the reflecting beam direction of the LIS, the determination unit **101** may determine a corresponding LIS and reflecting beam direction based on the time-frequency resource location.
(72) After the communication unit **102** acquires the above information, the determination unit **101** is configured to determine a specific LIS-assisted communication mode based on the measurement result, for example, to determine one or more LISs to be used in the LIS-assisted communication for the target UE and the reflecting beam direction of each LIS. The determination unit **101** is further configured to determine an associated user set of the target UE based on the measurement result. The determined one or more LISs also serve the UE in the associated user set. In this way, multiple LISs-based LIS-assisted cooperative transmission among multiple UEs may be realized to avoid interferences among UEs.
(73) For example, the determination unit **101** may be configured to take the reflecting beam direction of each LIS reported by the target UE as the reflecting beam direction of the LIS to be used in the LIS-assisted communication, and take the UE that receives signal of strength meeting the communication quality requirement in the reflecting beam direction of the LIS as the UE in the associated user set of the target UE, where the LIS also serves the UE in the associated user set.
(74) An example in which the determination unit **101** determines a specific LIS-assisted communication mode is given below with reference to FIG. **24**. FIG. **24** is a schematic diagram showing a scenario for LIS 1 and LIS 2 and UEs within a coverage range of the LISs. UE k is a target UE, an available LIS set includes LIS 1 and LIS 2, and an available serving UE set includes UE1-UE5 and UE k. After providing reference signal configuration information of the LIS 1 and LIS 2 to the UE, the base station transmits the LIS reference signal to the UE through a corresponding LIS. The UE measures LIS reference signal in each reflecting beam direction of each LIS and reports a measurement result in the above manner. UE 4 and UE 5 do not detect the reflecting beam meeting the communication quality requirement of the UE 4 and UE 5, so they do not report a measurement result.
(75) The measurement result reported by the target UE k and the remaining UEs 1-3 are shown in a table below. Table 1 shows the measurement result of these UEs for each reflecting beam direction of LIS 1. Table 2 shows the measurement result of these UEs for each reflecting beam direction of LIS 2.
(76) TABLE-US-00001 TABLE 1 LIS 1 beam 1 beam 2 beam 3 beam 4 beam 5 beam 6 UE 1 √ √ UE 2 √ √ √ UE 3 √ √ UE k √
(77) TABLE-US-00002 TABLE 2 LIS 2 beam 1 beam 2 beam 3 beam 4 beam 5 beam 6 UE 1 √ √ √ UE 2 √ √ √ UE 3 √ √ UE k √
(78) As shown in Table 1, an optimal reflecting beam direction of LIS 1 fed back from the target UE is beam 2. Therefore, the determination unit **101** determines the beam 2 of LIS 1 as the reflecting beam direction to be used. Meanwhile, the reflecting beam direction of LIS 1 meeting the communication quality requirement that is fed back from UE 2 and UE 3 also includes the beam 2. Therefore, the determination unit **101** determines UE2, UE3 and UE k as an associated user set of LIS 1, and LIS 1 serves the UEs in the associated user set.
(79) Similarly, as shown in Table 2, the optimal reflecting beam direction of LIS 2 fed back from the target UE is beam 5. Therefore, the determination unit **101** determines beam 5 of LIS 2 as the reflecting beam direction to be used. The reflecting beam direction of LIS 2 meeting the communication quality requirement that is fed back from UE 1 also includes the beam 5. Therefore, the determination unit **101** determines UE1 and UE k as an associated user set of LIS 2, and LIS 2 serves the UEs in the associated user set.

(80) In this way, LIS 1 serves UE2, UE3 and UE k, and LIS 2 serves UE1 and UE k, thus realizing a cooperative transmission mode in which multiple LISs serve multiple UEs.
(81) Next, the LIS-assisted communication transmission is performed between the base station and UE. For example, the base station transmits a directional beam to the determined LIS set, and the communication unit **102** is also configured to provide control information to one or more LISs in the LIS set, to cause each LIS to reflect an incident beam in the determined reflecting beam direction. That is, the base station controls a reflection phase of each LIS through the control information.
(82) In summary, the electronic apparatus **100** according to this embodiment configures a specific LIS reference signal for the base station and UE under the LIS-assisted communication mode, so as to achieve accurate channel measurement under the LIS-assisted communication mode, thus realizing the cooperative transmission mode for multiple LISs and multiple UEs, to avoid or alleviate interferences among UEs. By means of a passive device like LIS, the overhead can be reduced, and energy-saving and green communication can be realized. In addition, the problem of a communication blind spot in a cell is solved with the LIS-assisted communication.
Second Embodiment
(83) FIG. **25** is a block diagram showing functional modules of an electronic apparatus **200** according to another embodiment of the present disclosure. As shown in FIG. **25**, the electronic apparatus **200** includes a communication unit **201** and a measurement unit **202**. The communication unit **201** is configured to acquire, from a base station, configuration information of an LIS reference signal to be used for channel status measurement under an LIS-assisted communication mode. The measurement unit **202** is configured to perform measurement of the LIS reference signal based on the configuration information.
(84) The communication unit **201** and the measurement unit **202** may be implemented by one or more processing circuits. The processing circuits may be implemented as a chip and a processor, for example. Furthermore, it should be understood that functional units in the electronic apparatus shown in FIG. **25** are only logical modules divided according to the implemented specific functions, and are not intend to limit the specific implementations.
(85) The electronic apparatus **200** may be arranged on a UE side or communicatively connected to the UE, for example. Here, it should also be noted that the electronic apparatus **200** may be implemented in a chip level, or in an apparatus level. For example, the electronic apparatus **200** may function as the UE itself, and may also include an external device such as a memory, a transceiver (not shown). The memory may be used to store programs required by the user equipment to achieve various functions and related data information. The transceiver may include one or more communication interfaces to support communication with different devices (e.g., base station, other user equipment, or the like). Implementations of the transceiver are not limited here.
(86) As described in the first embodiment, the base station may determine that the target UE is to apply the LIS-assisted communication mode, or the target UE may trigger the application of the LIS-assisted communication mode. In the following, a UE that is determined to apply the LIS-assisted communication mode or requests to apply the LIS-assisted communication mode is called the target UE for the purpose of distinguishing.
(87) In a case of being determined to apply the LIS-assisted communication mode, the communication unit **201** may be configured to transmit communication quality information of the target UE, such as CQI, to a base station, so that the base station determines that the target UE is to apply the LIS-assisted communication mode in a case that the communication quality of the target UE drops below a predetermined threshold. In addition, the communication unit **201** may also transmit a priority level of the target UE to the base station, so that the base station may also take the priority level of the target UE into account when determining whether the target UE is to apply the LIS-assisted communication mode. For example, when the priority level of the target UE is high, the base station increases the possibility of applying the LIS-assisted communication mode

for the target UE.

(88) In a case of requesting to apply the LIS-assisted communication mode, for example, the communication unit **201** may further be configured to transmit an LIS assisting request to the base station in a case that the communication quality of the target UE is lower than a predetermined threshold, to indicate to the base station that the target UE requests to apply the LIS-assisted communication mode.

(89) The communication quality of the target UE may be measured by various QoS indexes, such as RSRP, RSRQ, SINR.

(90) As described in the first embodiment, the base station determines, in the case of determining the target UE is to apply the LIS-assisted communication mode, an LIS set to be used for LIS-assisted communication for the target UE and an available serving UE set for each LIS in the LIS set. For example, the LIS set includes respective LISs within whose coverage range the target UE is located, and the available serving UE set of an LIS includes UEs within a coverage range of the LIS. In an example, the communication unit **201** is also configured to transmit, to the base station, an identifier of an LIS within whose coverage range the target UE is located, so that the base station may determine the LIS set. Of course, the base station may also determine the LIS set and the available serving UE set described above according to its pre-stored location information of the LIS and location information of each UE. The specific details have been given in the first embodiment, and will not be repeated here.

(91) The communication unit **201** acquires configuration information of LIS reference signals related to LIS in the LIS set from the base station, for example, through RRC signaling. This configuration information is used to know, for example, how the UE receives and parses LIS reference signals, how to report a measurement result, etc.

(92) The measurement unit **202** performs measurement of the LIS reference signal and reports a measurement result according to the configuration information.

(93) For example, the configuration information of the LIS reference signal may include one or more of the following: a time-frequency resource location where the LIS reference signal is located, a correspondence relationship between LIS reference signals on different time-frequency locations and LISs, periodical reporting or non-periodical reporting of a measurement result.

(94) With this configuration information, the UE may know on which time-frequency resources the LIS reference signal is received, which LIS the received LIS reference signal comes from, and further which reflecting beam direction of the LIS the received LIS reference signal comes from, and whether to perform periodical reporting or non-periodical reporting of a measurement result. In other words, the correspondence relationship between LIS reference signals on different time-frequency resource locations and LISs may also include a correspondence relationship between LIS reference signals on different time-frequency resource locations and reflecting beam directions of different LISs in more detail.

(95) For example, in response to the configuration information, the measurement unit **202** may periodically measure the LIS reference signal, and the communication unit **201** periodically reports the measurement result of the LIS reference signal to the base station.

(96) In addition, the communication unit **201** may also acquire trigger signaling for reporting the measurement result of the LIS reference signal from the base station. The measurement unit **202** measures the LIS reference signal aperiodically in response to the trigger signaling, and the communication unit **201** reports an aperiodic measurement result of the LIS reference signal.

(97) For example, LIS reference signals may be configured as follows: LIS reference signals for different LISs are distinguished in frequency domain, and LIS reference signals for different reflecting beam directions of the same LIS are distinguished in time domain. That is, LIS reference signals for different LISs occupy different frequency domain resources, and LIS reference signals for different reflecting beam directions of the same LIS occupy different time domain parts of the frequency domain resources of the LIS.

(98) For example, the reference signals for different LISs are distributed on different subcarriers of OFDM symbols, and the LIS reference signals for different reflecting beam directions of the same LIS are distributed on different resource elements (RE) of the same subcarrier of the OFDM.
(99) In an example, the LIS reference signal is non-precoded/non-beamformed reference signal, and the LIS reference signals for different LISs do not overlap in time domain. In this case, since the LIS reference signal is non-directional and may be reflected by multiple LISs at the same time, the LIS reference signals for different LISs should be assigned with different durations, so that the UE may distinguish the LIS reference signals from different LISs.
(100) In another example, the LIS reference signal is a precoded/beamformed reference signal, and the LIS reference signals for different LISs overlap in the time domain. Because the LIS reference signals are digitally pre-coded, each OFDM symbol may have different subcarriers. The base station may transmit corresponding LIS reference signals directionally according to a position of each LIS. The UE may distinguish LIS reference signals from different LISs according to the subcarriers. In other words, in this case, the LIS reference signals for different LISs may overlap in the time domain, thus having higher configuration flexibility.
(101) In the first embodiment, the configuration mode of LIS reference signals has been described in detail with reference to FIGS. **9** to **23**, and will not be repeated here.
(102) The base station transmits the LIS reference signal according to the configuration information of the LIS reference signal, and meanwhile, the base station sequentially controls the reflecting beam direction of each LIS to obtain the desired reflecting beam for downlink beam scanning of the base station-LIS-UE. The UE sequentially measures received signal strength of a corresponding LIS reference signal, and feeds a measurement result back to the base station. The base station selects an appropriate channel for the UE based on the measurement result to perform the LIS-assisted communication.
(103) As an example, in a case that UE is the target UE, the measurement unit **202** is configured to measure the received signal strength of LIS reference signals corresponding to different reflecting beam directions of different LISs, determine, for each LIS, the reflecting beam direction corresponding to the maximum received signal strength; and the communication unit **201** reports the LIS to the base station in association with determined information of reflecting beam direction.
(104) In a case that UE is not the target UE, such as the remaining UE in an available serving UE set, the measurement unit **202** is configured to measure the received signal strength of LIS reference signals corresponding to different reflecting beam directions of different LISs, determine the reflecting beam direction of an LIS meeting the communication quality requirement of the UE, and the communication unit **201** reports the LIS to the base station in association with the determined information of the reflecting beam direction.
(105) It can be understood that if none of reflecting beams of an LIS can meet the communication quality requirement of the UE, the UE may not report for the LIS. If none of reflecting beams of all of the LISs can meet the communication quality requirement of the UE, the UE may continue to maintain direct communication with the base station without reporting a measurement result.
(106) For example, the measurement result reported by the communication unit **201** may include an identifier (ID) of the LIS and an ID of the reflecting beam direction. For example, for the target UE, the measurement result represents an optimal reflecting beam direction of LIS in the LIS set. For the remaining UE, the measurement result represents an LIS and a reflecting beam direction of the LIS that meet the communication quality requirement of the remaining UE.
(107) Alternatively, the measurement result reported by the UE may include a time-frequency resource location of the LIS reference signal. Since the time-frequency resource location of the LIS reference signal corresponds to the reflecting beam direction of the LIS, the base station may determine a corresponding LIS and reflecting beam direction based on the time-frequency resource location.
(108) After the base station acquires the above information, the base station determines a specific

LIS-assisted communication mode based on the measurement result, for example, determines one or more LISs to be used in the LIS-assisted communication for the target UE and the reflecting beam direction of each LIS. The base station further determines an associated user set of the target UE based on the measurement result, where the determined one or more LISs also serve the UEs in the associated user set. In this way, multiple LISs-based LIS assisted cooperative transmission among multiple UEs may be realized to avoid interferences among UEs.

(109) Finally, not only the target UE, but also all UEs in the associated user set perform LIS-assisted transmission to avoid interferences among UEs. It should be understood that the above description does not exclude the case where only the target UE is included in the associated user set.

(110) In summary, the electronic apparatus **200** according to this embodiment acquires a specific LIS reference signal configured for a base station and UE under the LIS-assisted communication mode, so as to achieve accurate channel measurement under the LIS-assisted communication mode, thus realizing the cooperative transmission mode for multiple LISs and multiple UEs, to avoid or alleviate interferences among UEs. By using a passive device like LIS, the overhead can be reduced, and energy-saving and green communication can be realized. In addition, the problem of a communication blind spot in a cell is solved with the LIS-assisted communication.

(111) For ease of understanding, FIG. **26** is a schematic diagram showing an example of information procedure among a base station (gNB), LIS and UE. Note that the UE here represents a set of a target UE and the remaining UEs. First, the target UE detects that the QoS of the target UE drops below a predetermined threshold, and thus transmits an LIS assisting request to gNB. The gNB determines to start LIS-assisted communication after receiving the LIS assisting request. The gNB determines an LIS set for the target UE and an available serving UE set according to location information of the target UE and the remaining UEs and pre-stored location information of the LISs, and transmits configuration information of the LIS reference signal to the UEs in the available serving UE set. In addition, in the case of non-periodical reporting, the base station also transmits the trigger signaling for reporting a measurement result to the UE. Next, the gNB transmits a directional beam carrying the LIS reference signal to the LIS according to the configuration information and controls a reflecting beam direction of the LIS through control information. After receiving a passive reflecting beam from LIS, the UE performs measurement and determines an LIS ID and beam ID to be reported based on a measurement result. It should be understood that a corresponding time-frequency resource location may also be reported here. As mentioned above, the measurement result reported by the target UE indicates an optimal reflecting beam direction of each LIS for the target UE, and the measurement result reported by the remaining UE indicates a reflecting beam direction of LIS meeting the communication quality requirement of the remaining UE. The base station determines a specific assisted transmission mode based on the obtained measured result, that is, determines which reflecting beam direction of an LIS is used, and an associated user set for which the LISs provide services at the same time. Then, the base station provides the LIS-assisted communication for UEs in the associated user set with this specific assisted transmission mode, which is realized by signal transmission to related LISs and reflecting beam control.

(112) It should be understood that FIG. **26** is only an example and not restrictive.

The Third Embodiment

(113) In the above description of embodiments of the electronic apparatuses for wireless communications, it is apparent that some processing and methods are further disclosed. In the following, a summary of the methods are described without repeating details that are described above. However, it should be noted that although the methods are disclosed when describing the electronic apparatuses for wireless communications, the methods are unnecessary to adopt those components or to be performed by those components described above. For example, implementations of the electronic apparatuses for wireless communications may be partially or

completely implemented by hardware and/or firmware. Methods for wireless communications to be discussed blow may be completely implemented by computer executable programs, although these methods may be implemented by the hardware and/or firmware for implementing the electronic apparatuses for wireless communications.
(114) FIG. **27***a* shows a flowchart of a method for wireless communications according to an embodiment of the present disclosure. The method includes: determining whether a target UE is to apply an LIS-assisted communication mode (S**11**); and in a case of determining the target UE is to apply the LIS-assisted communication mode, transmitting, to the target UE, configuration information of an LIS reference signal to be used for channel status measurement under the LIS-assisted communication mode (S**12**). The method may be performed on a base station side, for example.
(115) For example, in a case that a communication quality of the target UE is lower than a predetermined threshold, it is determined that the target UE is to apply the LIS-assisted communication mode. Alternatively, in response to an LIS assisting request from the target UE, it is determined that the target UE is to apply the LIS-assisted communication mode.
(116) The above step S**12**, may be for example, implemented as follows: in a case of determining the target UE is to apply the LIS-assisted communication mode, an LIS set to be used for LIS-assisted communication for the target UE and an available serving UE set for each LIS in the LIS set are determined, and configuration information of the LIS reference signal related to the LIS in the LIS set is transmitted to each of UE in the available serving UE set, where the LIS set includes respective LISs within whose coverage range the target UE is located, and the available serving UE set of an LIS includes UE within a coverage range of the LIS.
(117) For example, the LIS set and the available serving UE set may be determined based on pre-stored location information of the LIS and location information of each UE.
(118) In addition, one or more of the following may be acquired from the target UE: communication quality of the target UE, an identifier of the LIS within whose coverage range the target UE is located, and a priority level of the target UE.
(119) As shown in the dashed line block in FIG. **27***a*, the method also includes: transmitting the LIS reference signal to each UE via a corresponding LIS according to the configuration information of the LIS reference signal (S**13**), and acquiring a measurement result reported by each UE for measurement of the LIS reference signal (S**14**).
(120) The LIS reference signal may be transmitted periodically and the measurement result of the LIS reference signal may be acquired periodically. Trigger signaling for reporting the measurement result of LIS reference signal may also be transmitted to each UE, and an aperiodic measurement result of the LIS reference signal may be acquired.
(121) The configuration information of the LIS reference signal may include one or more of the following: a time-frequency resource location where the LIS reference signal is located, a correspondence relationship between LIS reference signals on different time-frequency locations and LISs, periodical reporting or non-periodical reporting of a measurement result. For example, LIS reference signals for different LISs may be distinguished in frequency domain, and LIS reference signals for different reflecting beam directions of the same LIS are distinguished in time domain. For example, the LIS reference signals for different LISs are distributed on different subcarriers of OFDM symbols, and the LIS reference signals for different reflecting beam directions of the same LIS are distributed on different resource elements of the same subcarrier of OFDM symbols.
(122) The LIS reference signal may be non-precoded/non-beamformed reference signal, and the LIS reference signals for different LISs do not overlap in time domain. The LIS reference signal may also be precoded/beamformed reference signal and the LIS reference signals for different LISs overlap in the time domain.
(123) In step S**14**, the following information may be acquired from the target UE: with respect to

each LIS in the LIS set, information of a reflecting beam direction in which the target UE receives signal of the maximum strength from the LIS. The following information may be acquired from each UE among at least a part of remaining UEs in the available serving UE set other than the target UE: with respect to each LIS in the LIS set, information of a reflecting beam direction in which the UE receives signal of strength meeting a communication quality requirement.

(124) For example, the reflecting beam direction of each LIS reported by the target UE may be used as the reflecting beam direction of the LIS to be used in LIS-assisted communication, and the UE that receives signal of strength meeting a communication quality requirement in the reflecting beam direction of the LIS may be used as a UE in an associated user set of the target UE, where the LIS also serves the UE in the associated user set.

(125) The measurement result includes, for example, an identifier of the LIS and an identifier of the reflecting beam direction, or a time-frequency resource location of the LIS reference signal.

(126) The method also includes a step S**15**: determining one or more LISs and the reflecting beam direction of each LIS to be used in LIS-assisted communication of the target UE based on the measurement result, and determining an associated user set of the target UE based on the measurement result, where one or more LISs also serve the UEs in the associated user set.

(127) The method also includes providing control information to one or more LISs so that each LIS reflects an incident beam in the determined reflecting beam direction.

(128) FIG. **27***b* shows a flowchart of a method for wireless communications according to another embodiment of the present disclosure. The method includes: acquiring, from a base station, configuration information of an LIS reference signal to be used for channel status measurement under an LIS-assisted communication mode (S**22**); and performing measurement of the LIS reference signal based on the configuration information (S**23**). The method may be performed on a UE side, for example.

(129) As shown in the dashed line block in FIG. **27***b*, the method may also include a step S**21**: in a case that communication quality of the target UE is lower than a predetermined threshold, transmitting an LIS assisting request to the base station to indicate to the base station that the target UE requests to apply the LIS-assisted communication mode.

(130) In addition, one or more of the following may be transmitted to the base station: communication quality of the target UE, an identifier of the LIS within whose coverage range the target UE is located, and a priority level of the target UE.

(131) The configuration information of the LIS reference signal may include one or more of the following: a time-frequency resource location where the LIS reference signal is located, a correspondence relationship between LIS reference signals on different time-frequency locations and LISs, periodical reporting or non-periodical reporting of a measurement result.

(132) The method may also include a step S**24**: reporting a measurement result to the base station.

(133) In step S**23**, the LIS reference signal may be measured periodically, and the measurement result of the LIS reference signal may be periodically reported to the base station in S**24**. Alternatively, trigger signaling for reporting the measurement result of the LIS reference signal may also be acquired from the base station, the LIS reference signal may be measured aperiodically in response to the trigger signaling, and an aperiodic measurement result of the LIS reference signal may be report in S**24**.

(134) For example, LIS reference signals for different LISs may be distinguished in frequency domain, and LIS reference signals for different reflecting beam directions of the same LIS may be distinguished in time domain. For example, the LIS reference signals for different LISs are distributed on different subcarriers of OFDM symbols, and the LIS reference signals for different reflecting beam directions of the same LIS are distributed on different resource elements of the same subcarrier of OFDM symbols.

(135) The LIS reference signal may be non-precoded/non-beamformed reference signal, and the LIS reference signals for different LISs do not overlap in the time domain. The LIS reference signal

may also be precoded/beamformed reference signal and the LIS reference signals for different LISs overlap in the time domain.

(136) For the target UE, received signal strength of the LIS reference signals corresponding to different reflecting beam directions of different LISs is measured in step S**23**, the reflecting beam direction corresponding to the maximum received signal strength is determined for each LIS, and the LIS is reported to the base station in association with the determined information of reflecting beam direction in step S**24**.

(137) For the remaining UE, received signal strength of LIS reference signals corresponding to different reflecting beam directions of different LISs is measured in step S**23**, a reflecting beam direction of an LIS meeting communication quality requirement of the UE is determined, and the LIS is reported to the base station in association with information of the reflecting beam direction in step S**24**.

(138) The measurement result includes, for example, an identifier of the LIS and an identifier of the reflecting beam direction, or a time-frequency resource location of the LIS reference signal.

(139) Note that the methods described above may be used in combination or separately. The details have been described in detail in the first to second embodiments, and will not be repeated here.

(140) The technology according to the present disclosure may be applied to various products.

(141) For example, the electronic apparatus **100** may be implemented as various base stations. The base stations may be implemented as any type of evolved node B (eNB) or gNB (5G base station). The eNB includes a macro eNB and a small eNB, for example. The small eNB may be an eNB such as a pico eNB, a micro eNB and a home (femto) eNB that covers a cell smaller than a macro cell. The situation is similar to the gNB. Alternatively, the base station may also be implemented as a base station of any other type, such as a NodeB and a base transceiver station (BTS). The base station may include a main body (that is also referred to as a base station device) configured to control wireless communications, and one or more remote radio heads (RRH) arranged in a different place from the main body. In addition, various types of user equipment each may operate as the base station by performing functions of the base station temporarily or semi-permanently.

(142) The electronic apparatus **200** may be implemented as as various types of user equipment. The user equipment may be implemented as a mobile terminal (such as a smartphone, a tablet personal computer (PC), a notebook PC, a portable game terminal, a portable/dongle type mobile router, and a digital camera), or an in-vehicle terminal (such as a car navigation device). The user equipment may also be implemented as a terminal (that is also referred to as a machine type communication (MTC) terminal) that performs machine-to-machine (M2M) communication. Furthermore, the user equipment may be a wireless communication module (such as an integrated circuit module including a single die) mounted on each of the terminals.

APPLICATION EXAMPLES REGARDING A BASE STATION

First Application Example

(143) FIG. **28** is a block diagram showing a first example of an exemplary configuration of an eNB or gNB to which the technology according to the present disclosure may be applied. It should be noted that the following description is given by taking the eNB as an example, which is also applicable to the gNB. An eNB **800** includes one or more antennas **810** and a base station apparatus **820**. The base station apparatus **820** and each of the antennas **810** may be connected to each other via a radio frequency (RF) cable.

(144) Each of the antennas **810** includes a single or multiple antennal elements (such as multiple antenna elements included in a multiple-input multiple-output (MIMO) antenna), and is used for the base station apparatus **820** to transmit and receive wireless signals. As shown in FIG. **28**, the eNB **800** may include the multiple antennas **810**. For example, the multiple antennas **810** may be compatible with multiple frequency bands used by the eNB **800**. Although FIG. **28** shows the example in which the eNB **800** includes the multiple antennas **810**, the eNB **800** may include a single antenna **810**.

(145) The base station apparatus **820** includes a controller **821**, a memory **822**, a network interface **823**, and a radio communication interface **825**.
(146) The controller **821** may be, for example, a CPU or a DSP, and operates various functions of a higher layer of the base station apparatus **820**. For example, the controller **821** generates a data packet from data in signals processed by the radio communication interface **825**, and transfers the generated packet via the network interface **823**. The controller **821** may bundle data from multiple base band processors to generate the bundled packet, and transfer the generated bundled packet. The controller **821** may have logical functions of performing control such as resource control, radio bearer control, mobility management, admission control and scheduling. The control may be performed in corporation with an eNB or a core network node in the vicinity. The memory **822** includes a RAM and a ROM, and stores a program executed by the controller **821** and various types of control data (such as a terminal list, transmission power data and scheduling data).
(147) The network interface **823** is a communication interface for connecting the base station apparatus **820** to a core network **824**. The controller **821** may communicate with a core network node or another eNB via the network interface **823**. In this case, the eNB **800**, and the core network node or another eNB may be connected to each other via a logic interface (such as an S1 interface and an X2 interface). The network interface **823** may also be a wired communication interface or a wireless communication interface for wireless backhaul. In a case that the network interface **823** is a wireless communication interface, the network interface **823** may use a higher frequency band for wireless communication than that used by the radio communication interface **825**.
(148) The radio communication interface **825** supports any cellular communication scheme (such as Long Term Evolution (LTE) and LTE-advanced), and provides wireless connection to a terminal located in a cell of the eNB **800** via the antenna **810**. The radio communication interface **825** may typically include, for example, a baseband (BB) processor **826** and an RF circuit **827**. The BB processor **826** may perform, for example, encoding/decoding, modulating/demodulating, and multiplexing/demultiplexing, and perform various types of signal processing of layers (such as L1, Media Access Control (MAC), Radio Link Control (RLC), and a Packet Data Convergence Protocol (PDCP)). The BB processor **826** may have a part or all of the above-described logical functions, to replace the controller **821**. The BB processor **826** may be a memory storing communication control programs, or a module including a processor and a related circuit configured to execute the programs. Updating the program may allow the functions of the BB processor **826** to be changed. The module may be a card or a blade inserted into a slot of the base station apparatus **820**. Alternatively, the module may be a chip mounted on the card or the blade. Meanwhile, the RF circuit **827** may include, for example, a mixer, a filter, and an amplifier, and transmits and receives wireless signals via the antenna **810**.
(149) As shown in FIG. **28**, the radio communication interface **825** may include multiple BB processors **826**. For example, the multiple BB processors **826** may be compatible with multiple frequency bands used by the eNB **800**. The radio communication interface **825** may include multiple RF circuits **827**, as shown in FIG. **28**. For example, the multiple RF circuits **827** may be compatible with multiple antenna elements. Although FIG. **28** shows the example in which the radio communication interface **825** includes multiple BB processors **826** and multiple RF circuits **827**, the radio communication interface **825** may include a single BB processor **826** and a single RF circuit **827**.
(150) In the eNB **800** as shown in FIG. **28**, the communication unit **102** and the transceiver of the electronic apparatus **100** may be implemented by the radio communication interface **825**. At least some of the functions may also be implemented by the controller **821**. For example, the controller **821** may configure the LIS reference signal for the UE, perform measurement of the LIS reference signal and report a measurement result, and determine a specific LIS-assisted communication mode based on the measurement result to achieve the cooperative LIS-assisted communication among multiple UEs, by performing the functions of the determination unit **101** and the communication

unit **102**.

Second Application Example

(151) FIG. **29** is a block diagram showing a second example of a schematic configuration of an eNB or gNB to which the technology of the present disclosure can be applied. It should be noted that the following description is given by taking the eNB as an example, which is also applied to the gNB. An eNB **830** includes one or more antennas **840**, a base station apparatus **850**, and an RRH **860**. The RRH **860** and each of the antennas **840** may be connected to each other via an RF cable. The base station apparatus **850** and the RRH **860** may be connected to each other via a high speed line such as an optical fiber cable.

(152) Each of the antennas **840** includes a single or multiple antennal elements (such as multiple antenna elements included in an MIMO antenna), and is used for the RRH **860** to transmit and receive wireless signals. As shown in FIG. **29**, the eNB **830** may include multiple antennas **840**. For example, the multiple antennas **840** may be compatible with multiple frequency bands used by the eNB **830**. Although FIG. **29** shows the example in which the eNB **830** includes multiple antennas **840**, the eNB **830** may include a single antenna **840**.

(153) The base station apparatus **850** includes a controller **851**, a memory **852**, a network interface **853**, a radio communication interface **855**, and a connection interface **857**. The controller **851**, the memory **852**, and the network interface **853** are the same as the controller **821**, the memory **822**, and the network interface **823** described with reference to FIG. **28**.

(154) The radio communication interface **855** supports any cellular communication scheme (such as LTE and LTE-advanced), and provides wireless communication to a terminal located in a sector corresponding to the RRH **860** via the RRH **860** and the antenna **840**. The radio communication interface **855** may typically include, for example, a BB processor **856**. The BB processor **856** is the same as the BB processor **826** described with reference to FIG. **28**, except that the BB processor **856** is connected to an RF circuit **864** of the RRH **860** via the connection interface **857**. As show in FIG. **29**, the radio communication interface **855** may include multiple BB processors **856**. For example, the multiple BB processors **856** may be compatible with multiple frequency bands used by the eNB **830**. Although FIG. **29** shows the example in which the radio communication interface **855** includes multiple BB processors **856**, the radio communication interface **855** may include a single BB processor **856**.

(155) The connection interface **857** is an interface for connecting the base station apparatus **850** (radio communication interface **855**) to the RRH **860**. The connection interface **857** may also be a communication module for communication in the above-described high speed line that connects the base station apparatus **850** (radio communication interface **855**) to the RRH **860**.

(156) The RRH **860** includes a connection interface **861** and a radio communication interface **863**.

(157) The connection interface **861** is an interface for connecting the RRH **860** (radio communication interface **863**) to the base station apparatus **850**. The connection interface **861** may also be a communication module for communication in the above-described high speed line.

(158) The radio communication interface **863** transmits and receives wireless signals via the antenna **840**. The radio communication interface **863** may typically include, for example, an RF circuit **864**. The RF circuit **864** may include, for example, a mixer, a filter and an amplifier, and transmits and receives wireless signals via the antenna **840**. The radio communication interface **863** may include multiple RF circuits **864**, as shown in FIG. **29**. For example, the multiple RF circuits **864** may support multiple antenna elements. Although FIG. **29** shows the example in which the radio communication interface **863** includes multiple RF circuits **864**, the radio communication interface **863** may include a single RF circuit **864**.

(159) In the eNB **830** shown in FIG. **29**, the communication unit **102** and the transceiver of the electronic apparatus **100** may be implemented by the radio communication interface **855** and/or radio communication interface **863**. At least some of the functions may also be implemented by the controller **851**. For example, the controller **851** may configure the LIS reference signal for the UE,

perform measurement of the LIS reference signal and report a measurement result, and determine a specific LIS-assisted communication mode based on the measurement result, to achieve the cooperative LIS-assisted communication among multiple UEs by performing the functions of the determination unit **101** and the communication unit **102**.

APPLICATION EXAMPLE REGARDING USER EQUIPMENT

First Application Example

(160) FIG. **30** is a block diagram showing an exemplary configuration of a smartphone **900** to which the technology according to the present disclosure may be applied. The smartphone **900** includes a processor **901**, a memory **902**, a storage **903**, an external connection interface **904**, a camera **906**, a sensor **907**, a microphone **908**, an input device **909**, a display device **910**, a speaker **911**, a radio communication interface **912**, one or more antenna switches **915**, one or more antennas **916**, a bus **917**, a battery **918**, and an auxiliary controller **919**.

(161) The processor **901** may be, for example, a CPU or a system on a chip (SoC), and controls functions of an application layer and another layer of the smartphone **900**. The memory **902** includes a RAM and a ROM, and stores a program executed by the processor **901** and data. The storage **903** may include a storage medium such as a semiconductor memory and a hard disk. The external connection interface **904** is an interface for connecting an external device (such as a memory card and a universal serial bus (USB) device) to the smartphone **900**.

(162) The camera **906** includes an image sensor (such as a charge coupled device (CCD) and a complementary metal oxide semiconductor (CMOS)), and generates a captured image. The sensor **907** may include a group of sensors, such as a measurement sensor, a gyro sensor, a geomagnetism sensor, and an acceleration sensor. The microphone **908** converts sounds inputted to the smartphone **900** to audio signals. The input device **909** includes, for example, a touch sensor configured to detect touch onto a screen of the display device **910**, a keypad, a keyboard, a button, or a switch, and receives an operation or information inputted from a user. The display device **910** includes a screen (such as a liquid crystal display (LCD) and an organic light-emitting diode (OLED) display), and displays an output image of the smartphone **900**. The speaker **911** converts audio signals outputted from the smartphone **900** to sounds.

(163) The radio communication interface **912** supports any cellular communication scheme (such as LTE and LTE-advanced), and performs wireless communications. The radio communication interface **912** may include, for example, a BB processor **913** and an RF circuit **914**. The BB processor **913** may perform, for example, encoding/decoding, modulating/demodulating, and multiplexing/de-multiplexing, and perform various types of signal processing for wireless communication. The RF circuit **914** may include, for example, a mixer, a filter and an amplifier, and transmits and receives wireless signals via the antenna **916**. It should be noted that although FIG. **30** shows a case that one RF link is connected to one antenna, which is only illustrative, and a situation where one RF link is connected to multiple antennas through multiple phase shifters is also possible. The radio communication interface **912** may be a chip module having the BB processor **913** and the RF circuit **914** integrated thereon. The radio communication interface **912** may include multiple BB processors **913** and multiple RF circuits **914**, as shown in FIG. **30**. Although FIG. **30** shows the example in which the radio communication interface **912** includes multiple BB processors **913** and multiple RF circuits **914**, the radio communication interface **912** may include a single BB processor **913** or a single RF circuit **914**.

(164) Furthermore, in addition to a cellular communication scheme, the radio communication interface **912** may support another type of wireless communication scheme such as a short-distance wireless communication scheme, a near field communication scheme, and a wireless local area network (LAN) scheme. In this case, the radio communication interface **912** may include the BB processor **913** and the RF circuit **914** for each wireless communication scheme.

(165) Each of the antenna switches **915** switches connection destinations of the antennas **916** among multiple circuits (such as circuits for different wireless communication schemes) included

in the radio communication interface **912**.

(166) Each of the antennas **916** includes a single or multiple antenna elements (such as multiple antenna elements included in an MIMO antenna) and is used for the radio communication interface **912** to transmit and receive wireless signals. The smartphone **900** may include the multiple antennas **916**, as shown in FIG. **30**. Although FIG. **30** shows the example in which the smartphone **900** includes multiple antennas **916**, the smartphone **900** may include a single antenna **916**.

(167) Furthermore, the smartphone **900** may include the antenna **916** for each wireless communication scheme. In this case, the antenna switches **915** may be omitted from the configuration of the smartphone **900**.

(168) The bus **917** connects the processor **901**, the memory **902**, the storage **903**, the external connection interface **904**, the camera **906**, the sensor **907**, the microphone **908**, the input device **909**, the display device **910**, the speaker **911**, the radio communication interface **912**, and the auxiliary controller **919** to each other. The battery **918** supplies power to blocks of the smartphone **900** shown in FIG. **30** via feeder lines, which are partially shown as dashed lines in FIG. **30**. The auxiliary controller **919** operates a minimum necessary function of the smartphone **900**, for example, in a sleep mode.

(169) In the smart phone **900** as shown in FIG. **30**, the communication unit **102** and the transceiver of the electronic apparatus **200** may be implemented by the radio communication interface **912**. At least some of the functions may also be implemented by the processor **901** or auxiliary controller **919**. For example, the processor **901** or the auxiliary controller **919** may receive the LIS reference signal, perform the measurement of the LIS reference signal and report a measurement result, and achieve the cooperative LIS-assisted communication among multiple UEs by performing the functions of the communication unit **201** and the measurement unit **202**.

Second Application Example

(170) FIG. **31** is a block diagram showing an example of a schematic configuration of a car navigation apparatus **920** to which the technology according to the present disclosure may be applied. The car navigation apparatus **920** includes a processor **921**, a memory **922**, a global positioning system (GPS) module **924**, a sensor **925**, a data interface **926**, a content player **927**, a storage medium interface **928**, an input device **929**, a display device **930**, a speaker **931**, a radio communication interface **933**, one or more antenna switches **936**, one or more antennas **937**, and a battery **938**.

(171) The processor **921** may be, for example a CPU or a SoC, and controls a navigation function and additional function of the car navigation apparatus **920**. The memory **922** includes RAM and ROM, and stores a program executed by the processor **921**, and data.

(172) The GPS module **924** determines a position (such as latitude, longitude and altitude) of the car navigation apparatus **920** by using GPS signals received from a GPS satellite. The sensor **925** may include a group of sensors such as a gyro sensor, a geomagnetic sensor and an air pressure sensor. The data interface **926** is connected to, for example, an in-vehicle network **941** via a terminal that is not shown, and acquires data (such as vehicle speed data) generated by the vehicle.

(173) The content player **927** reproduces content stored in a storage medium (such as a CD and DVD) that is inserted into the storage medium interface **928**. The input device **929** includes, for example, a touch sensor configured to detect touch onto a screen of the display device **930**, a button, or a switch, and receives an operation or information inputted from a user. The display device **930** includes a screen such as an LCD or OLED display, and displays an image of the navigation function or reproduced content. The speaker **931** outputs a sound for the navigation function or the reproduced content.

(174) The radio communication interface **933** supports any cellular communication scheme (such as LTE and LTE-Advanced), and performs wireless communication. The radio communication interface **933** may typically include, for example, a BB processor **934** and an RF circuit **935**. The BB processor **934** may perform, for example, encoding/decoding, modulating/demodulating and

multiplexing/demultiplexing, and perform various types of signal processing for wireless communication. The RF circuit **935** may include, for example, a mixer, a filter and an amplifier, and transmits and receives wireless signals via the antenna **937**. The radio communication interface **933** may also be a chip module having the BB processor **934** and the RF circuit **935** integrated thereon. The radio communication interface **933** may include multiple BB processors **934** and multiple RF circuits **935**, as shown in FIG. **31**. Although FIG. **31** shows the example in which the radio communication interface **933** includes multiple BB processors **934** and multiple RF circuits **935**, the radio communication interface **933** may include a single BB processor **934** and a single RF circuit **935**.

(175) Furthermore, in addition to a cellular communication scheme, the radio communication interface **933** may support another type of wireless communication scheme such as a short-distance wireless communication scheme, a near field communication scheme, and a wireless LAN scheme. In this case, the radio communication interface **933** may include the BB processor **934** and the RF circuit **935** for each wireless communication scheme.

(176) Each of the antenna switches **936** switches connection destinations of the antennas **937** among multiple circuits (such as circuits for different wireless communication schemes) included in the radio communication interface **933**.

(177) Each of the antennas **937** includes a single or multiple antenna elements (such as multiple antenna elements included in an MIMO antenna), and is used for the radio communication interface **933** to transmit and receive wireless signals. As shown in FIG. **31**, the car navigation apparatus **920** may include multiple antennas **937**. Although FIG. **31** shows the example in which the car navigation apparatus **920** includes multiple antennas **937**, the car navigation apparatus **920** may include a single antenna **937**.

(178) Furthermore, the car navigation apparatus **920** may include the antenna **937** for each wireless communication scheme. In this case, the antenna switches **936** may be omitted from the configuration of the car navigation apparatus **920**.

(179) The battery **938** supplies power to the blocks of the car navigation apparatus **920** shown in FIG. **31** via feeder lines that are partially shown as dash lines in FIG. **31**. The battery **938** accumulates power supplied from the vehicle.

(180) In the car navigation equipment **920** as shown in FIG. **31**, the communication unit **201** and the transceiver of the electronic apparatus **200** may be implemented by the radio communication interface **933**. At least some of the functions may also be implemented by the processor **921**. For example, the processor **921** may receive the LIS reference signal, perform measurement of the LIS reference signal and report a measurement result, and achieve the cooperative LIS-assisted communication among multiple UEs by performing the functions of the communication unit **201** and the measurement unit **202**.

(181) The technology according to the present disclosure may also be implemented as an in-vehicle system (or a vehicle) **940** including one or more blocks of the car navigation device **920**, the in-vehicle network **941**, and a vehicle module **942**. The vehicle module **942** generates vehicle data (such as vehicle speed, engine speed, and failure information), and outputs the generated data to the in-vehicle network **941**.

(182) The basic principle of the present disclosure has been described above in conjunction with particular embodiments. However, as can be appreciated by those ordinarily skilled in the art, all or any of the steps or components of the method and apparatus according to the disclosure can be implemented with hardware, firmware, software or a combination thereof in any computing device (including a processor, a storage medium, etc.) or a network of computing devices by those ordinarily skilled in the art in light of the disclosure of the disclosure and making use of their general circuit designing knowledge or general programming skills.

(183) Moreover, the present disclosure further discloses a program product in which machine-readable instruction codes are stored. The aforementioned methods according to the embodiments

can be implemented when the instruction codes are read and executed by a machine.

(184) Accordingly, a memory medium for carrying the program product in which machine-readable instruction codes are stored is also covered in the present disclosure. The memory medium includes but is not limited to soft disc, optical disc, magnetic optical disc, memory card, memory stick and the like.

(185) In the case where the present disclosure is realized with software or firmware, a program constituting the software is installed in a computer with a dedicated hardware structure (e.g. the general computer **3200** shown in FIG. **32**) from a storage medium or network, wherein the computer is capable of implementing various functions when installed with various programs.

(186) In FIG. **32**, a central processing unit (CPU) **3201** executes various processing according to a program stored in a read-only memory (ROM) **3202** or a program loaded to a random access memory (RAM) **3203** from a memory section **3208**. The data needed for the various processing of the CPU **3201** may be stored in the RAM **3203** as needed. The CPU **3201**, the ROM **3202** and the RAM **3203** are linked with each other via a bus **3204**. An input/output interface **3205** is also linked to the bus **3204**.

(187) The following components are linked to the input/output interface **3205**: an input section **3206** (including keyboard, mouse and the like), an output section **3207** (including displays such as a cathode ray tube (CRT), a liquid crystal display (LCD), a loudspeaker and the like), a memory section **3208** (including hard disc and the like), and a communication section **3209** (including a network interface card such as a LAN card, modem and the like). The communication section **3209** performs communication processing via a network such as the Internet. A driver **3210** may also be linked to the input/output interface **3205**, if needed. If needed, a removable medium **3211**, for example, a magnetic disc, an optical disc, a magnetic optical disc, a semiconductor memory and the like, may be installed in the driver **3210**, so that the computer program read therefrom is installed in the memory section **3208** as appropriate.

(188) In the case where the foregoing series of processing is achieved through software, programs forming the software are installed from a network such as the Internet or a memory medium such as the removable medium **3211**.

(189) It should be appreciated by those skilled in the art that the memory medium is not limited to the removable medium **3211** shown in FIG. **32**, which has program stored therein and is distributed separately from the apparatus so as to provide the programs to users. The removable medium **3211** may be, for example, a magnetic disc (including floppy disc (registered trademark)), a compact disc (including compact disc read-only memory (CD-ROM) and digital versatile disc (DVD), a magneto optical disc (including mini disc (MD) (registered trademark)), and a semiconductor memory. Alternatively, the memory medium may be the hard discs included in ROM **3202** and the memory section **3208** in which programs are stored, and can be distributed to users along with the device in which they are incorporated.

(190) To be further noted, in the apparatus, method and system according to the present disclosure, the respective components or steps can be decomposed and/or recombined. These decompositions and/or re-combinations shall be regarded as equivalent solutions of the disclosure. Moreover, the above series of processing steps can naturally be performed temporally in the sequence as described above but will not be limited thereto, and some of the steps can be performed in parallel or independently from each other.

(191) Finally, to be further noted, the term “include”, “comprise” or any variant thereof is intended to encompass nonexclusive inclusion so that a process, method, article or device including a series of elements includes not only those elements but also other elements which have been not listed definitely or an element(s) inherent to the process, method, article or device. Moreover, the expression “comprising a (n) . . . ” in which an element is defined will not preclude presence of an additional identical element(s) in a process, method, article or device comprising the defined element(s)” unless further defined.

(192) Although the embodiments of the present disclosure have been described above in detail in connection with the drawings, it shall be appreciated that the embodiments as described above are merely illustrative rather than limitative of the present disclosure. Those skilled in the art can make various modifications and variations to the above embodiments without departing from the spirit and scope of the present disclosure. Therefore, the scope of the present disclosure is defined merely by the appended claims and their equivalents.

Claims

1. An electronic apparatus for wireless communications, comprising: processing circuitry, configured to: determine whether a target user equipment is to apply a large intelligent surface (LIS)-assisted communication mode; and in a case of determining the target user equipment is to apply the LIS-assisted communication mode, transmit, to the target user equipment, configuration information of an LIS reference signal to be used for channel status measurement under the LIS-assisted communication mode, wherein the processing circuitry is further configured to: determine, in the case of determining the target user equipment is to apply the LIS-assisted communication mode, an LIS set to be used for LIS-assisted communication for the target user equipment and an available serving user equipment set for each LIS in the LIS set, and transmit, to each of user equipment in the available serving user equipment set, configuration information of the LIS reference signal related to the LIS in the LIS set, wherein, the LIS set comprises respective LISs within whose coverage range the target user equipment is located, and the available serving user equipment set of an LIS comprises UEs within a coverage range of the LIS.
2. The electronic apparatus according to claim 1, wherein the processing circuitry is further configured to determine that the target user equipment is to apply the LIS-assisted communication mode in a case that a communication quality of the target user equipment is lower than a predetermined threshold; and/or wherein the processing circuitry is configured to determine that the target user equipment is to apply the LIS-assisted communication mode in response to an LIS assisting request from the target user equipment.
3. The electronic apparatus according to claim 1, wherein the processing circuitry is further configured to transmit the LIS reference signal to each of user equipment via a corresponding LIS according to the configuration information of the LIS reference signal, and acquire a measurement result reported by each of user equipment for measurement of the LIS reference signal.
4. The electronic apparatus according to claim 3, wherein the processing circuitry is configured to periodically transmit the LIS reference signal and periodically acquire a measurement result of the LIS reference signal; and/or wherein the processing circuitry is further configured to transmit, to each of user equipment, trigger signaling for reporting the measurement result of the LIS reference signal, and acquire an aperiodic measurement result of the LIS reference signal.
5. The electronic apparatus according to claim 1, wherein the configuration information of the LIS reference signal comprises one or more of the following: a time-frequency resource location where the LIS reference signal is located, a correspondence relationship between LIS reference signals on different time-frequency locations and LISs, periodical reporting or non-periodical reporting of a measurement result.
6. The electronic apparatus according to claim 1, wherein LIS reference signals for different LISs are distinguished in frequency domain, and LIS reference signals for different reflecting beam directions of the same LIS are distinguished in time domain.
7. The electronic apparatus according to claim 6, wherein LIS reference signals for different LISs are distributed on different subcarriers of orthogonal frequency division multiplexing (OFDM) symbols, and LIS reference signals for different reflecting beam directions of the same LIS are distributed on different resource elements of the same subcarrier of OFDM symbols.
8. The electronic apparatus according to claim 6, wherein the LIS reference signal is a non-

precoded/non-beamformed reference signal, and the LIS reference signals for different LISs do not overlap in the time domain; and/or wherein the LIS reference signal is a pre-coded/beamformed reference signal, and the LIS reference signals for different LISs overlap in the time domain.

9. The electronic apparatus according to claim 1, wherein the processing circuitry is further configured to acquire, from the target user equipment, one or more of the following: a communication quality of the target user equipment, an identifier of an LIS within whose coverage range the target user equipment is located, and a priority level of the target user equipment.

10. The electronic apparatus according to claim 1, wherein the processing circuitry is configured to determine the LIS set and the available serving user equipment set based on pre-stored location information of the LIS and location information of each of user equipment.

11. The electronic apparatus according to claim 3, wherein the processing circuitry is configured to acquire, from the target user equipment, with respect to each LIS in the LIS set, information of a reflecting beam direction in which the target user equipment receives signal of the maximum strength from the LIS; and wherein the processing circuitry is configured to acquire, from each of user equipment among at least a part of remaining UE in the available serving UE set other than the target user equipment, with respect to each LIS in the LIS set, information of a reflecting beam direction in which the user equipment receives signal of a strength meeting a communication quality requirement.

12. The electronic apparatus according to claim 3, wherein the measurement result comprises an identifier of an LIS and an identifier of a reflecting beam direction, or comprises a time-frequency resource location of an LIS reference signal.

13. The electronic apparatus according to claim 3, wherein the processing circuitry is configured to determine one or more LISs to be used in LIS-assisted communication for the target user equipment and the reflecting beam direction of each LIS based on the measurement result, and the processing circuitry is further configured to determine an associated user set of the target user equipment based on the measurement result, wherein the one or more LISs also serve user equipment in the associated user set.

14. The electronic apparatus according to claim 13, wherein the processing circuitry is configured to provide control information to the one or more LISs to cause each LIS to reflect an incident beam in a determined reflecting beam direction.

15. The electronic apparatus according to claim 11, wherein the processing circuitry is configured to take a reflecting beam direction of each LIS reported by the target user equipment as a reflecting beam direction of the LIS to be used in LIS-assisted communication, and take a user equipment that receives signal of strength meeting a communication quality requirement in the reflecting beam direction of the LIS as a user equipment in an associated user set of the target user equipment, wherein the LIS also serves user equipment in the associated user set.

16. An electronic apparatus for wireless communications, comprising: processing circuitry, configured to: acquire, from a base station, configuration information of an LIS reference signal to be used for channel status measurement under an LIS-assisted communication mode; and perform measurement of the LIS reference signal based on the configuration information, wherein LIS reference signals for different LISs are distinguished in frequency domain, and LIS reference signals for different reflecting beam directions of the same LIS are distinguished in time domain; wherein LIS reference signals for different LISs are distributed on different subcarriers of orthogonal frequency division multiplexing (OFDM) symbols, and LIS reference signals for different reflecting beam directions of the same LIS are distributed on different resource elements of the same subcarrier of OFDM symbols; wherein the LIS reference signal is a non-precoded/non-beamformed reference signal, and the LIS reference signals for different LISs do not overlap in the time domain; and wherein the LIS reference signal is a pre-coded/beamformed reference signal, and the LIS reference signals for different LISs overlap in the time domain.

17. The electronic apparatus according to claim 16, wherein the processing circuitry is further

configured to transmit an LIS assisting request to the base station in a case that a communication quality of a target user equipment is lower than a predetermined threshold, so as to indicate to the base station that the target user equipment requests to apply the LIS-assisted communication mode.

18. A method for wireless communications, comprising: determining whether a target user equipment is to apply a large intelligent surface (LIS)-assisted communication mode; and in a case of determining the target user equipment is to apply the LIS-assisted communication mode, transmitting, to the target user equipment, configuration information of an LIS reference signal to be used for channel status measurement under the LIS-assisted communication mode, wherein the method further comprises: determining, in the case of determining the target user equipment is to apply the LIS-assisted communication mode, an LIS set to be used for LIS-assisted communication for the target user equipment and an available serving user equipment set for each LIS in the LIS set, and transmitting, to each of user equipment in the available serving user equipment set, configuration information of the LIS reference signal related to the LIS in the LIS set, wherein, the LIS set comprises respective LISs within whose coverage range the target user equipment is located, and the available serving user equipment set of an LIS comprises UEs within a coverage range of the LIS.